

SHAGHAYEGH “SHAY” ROUHI

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SKILLS

Primary: Python, R, SQL | Machine Learning, Deep Learning, PyTorch, TensorFlow, scikit-learn, SciPy | Azure, Azure DevOps, Git, GitHub, CI/CD, Agile/Scrum | FastAPI, REST API | Microsoft SQL Server, SSIS, SSRS, SSAS, ETL | Power BI, A/B Testing, SharePoint, Visio

Additional: JavaScript, SAS | PySpark | Flask | Tableau | MongoDB, MariaDB | LLMs

EDUCATION

University of Nebraska-Lincoln, Lincoln, NE,

Bachelor of Science, with Distinction, May 2026 GPA: 3.89/4.0

Major: Data Science | **Minors:** Computer Science, Mathematics

RELEVANT EXPERIENCE

Data Scientist Intern, *Nebraska Department of Transportation (NDOT)*, Lincoln, NE | May 2024 – Present

- Engineer a sidewalk semantic segmentation model using a U-Net architecture with a ResNet-34 encoder to identify pedestrian pathways from geospatial imagery; collaborate with cross-functional GIS teams to integrate model outputs into an internal infrastructure mapping tool
- Develop a client facing financial scenario modeling tool in R Shiny to support executive stakeholders to evaluate funding scenarios, budget gaps, and support strategic resource planning decisions.
- Create interactive and automated dashboards using PowerBI, Shiny, and Tableau for multiple projects, to allow stakeholders measure snow coverage, budget allocation, and other performance metrics over time.
- Conduct a validation and cost analysis comparing AI traffic cameras and traditional sensors against manual counts using statistical methods such as Wilcoxon Signed-rank test to determine the most cost-efficient and accurate method for traffic count.

Undergraduate Research – Department of Mathematics, *University of Nebraska-Lincoln*, Lincoln, NE | September 2024 – December 2025

- Developed a nonlocal Deep Neural Network (DNN) training method by replacing classical derivatives with integral operators, to improve robustness and stability of the training.
- Validated the proposed nonlocal DNN by performing case studies on multiple datasets and DNN parameters in Python (NumPy, TensorFlow), demonstrating the effectiveness of nonlocal methods.
- Supported by funding from Undergraduate Creative Activities and Research Experience (UCARE) program (September 2024 – May 2025), resulting in a poster presentation at UNL Research Day.

PROJECTS

E-commerce Customer Churn Analysis & Dashboard

- Analyzed customer churn behavior in an e-commerce platform using SQL to develop strategic business decisions.
- Identified key churn trends and designed a Power BI dashboard to visualize churn patterns in dimensions such as tenure, delivery distance, and complaint frequency, allowing stakeholders to make informed decisions.

Resume-Job Matcher | [Live Demo](#)

- Developed an NLP based analytical tool to evaluate the alignment of resumes with job descriptions, emphasizing feature extraction and similarity scoring, producing match metrics and improvement suggestions.
- Implemented the application using Streamlit with CI/CD automation through GitHub Actions to support a more consistent and faster development cycle.

BananaBath, *Interior Design Assistant (Hackathon – CornHacks 2025)*

- Designed a full stack end-to-end interior design application through Python, Gemini API, Flask, and HTML/CSS/JavaScript under a 24-hour hackathon deadline.
- Integrated a mini RAG (Retrieval Augmented Generation) system that uses semantic vectors generated through a transformer model to retrieve specific design images from a database according to user input.

HONORS AND AWARDS

Dean's list, University of Nebraska-Lincoln (2022 –2026)

Merit-based Academic Scholarships (*University of Nebraska-Lincoln*)